Metric Geometry and Certified Periodic Images

Stage C0A2: immutable fit/analysis metrics and fail-closed triclinic closest-image geometry

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Scope

This specification owns the metric-bearing geometric contracts used by Stage C0A2. It does not own statistical kernel covariance, density topology, or site basins.

Three objects remain distinct:

- `RegistrationFitMetric`: geometry used only to fit reference-group translation;
- `AnalysisGeometryMetric`: geometry exposed to downstream periodic analysis;
- statistical kernel metrics, which remain outside this stage.

Neither Stage C0 metric is silently inherited from the other.

Row-vector metric convention

For a Cartesian row displacement δ,

$$\| \delta \|_P^2 = \delta P \delta^T,$$

where P is finite, symmetric, and positive definite. Under the fixed row-coordinate change $q' = qA$, the same geometry is represented by

$$P' = A^{-1} P A^{-T}.$$

Every metric records units, coordinate-frame meaning, transformation provenance, schema, and an immutable SHA-256 signature.

Certified triclinic closest image

For full-periodic cell G, solve

$$n_* = \arg\min_{n \in \mathbb{Z}^3} \| \delta - nG \|_P .$$

Componentwise fractional rounding is only a seed; it is not accepted as a proof for a skewed cell. Let $P = LL^T$, $B = GL$, and $f = \delta G^{-1}$. From any seed with residual radius r, every improving integer vector satisfies

$$\| f - n \|_2 \leq \frac{r}{\sigma_{\min}(B)}.$$

The implementation exhaustively enumerates the resulting finite integer box. This bound is derived directly from the smallest-singular-value inequality; no external approximate nearest-plane result is treated as the scientific solver. The result records the selected integer shift, closest vector, distance, second-best distance, separation, tie status, examined-candidate count, and certification status.

A search exceeding the declared candidate budget fails before returning an uncertified answer. Distances within the combined absolute/relative tie tolerance are ambiguous.

Acceptance requirements

- orthorhombic results agree with ordinary wrapping;
- skew cells may disagree with componentwise rounding, and the exhaustive result wins;
- a metric coordinate change preserves distance when both cell and metric are transformed;
- exact and near ties are reported rather than broken silently;
- serialization round trips preserve signatures;
- fit and analysis metrics remain independently configurable.